

DATA SCIENCE INTERNSHIP

Internship Completion Report

Submitted in partial fulfillment of the requirements for
B.Tech in Computer Science & Engineering (IBM)

Manthan Arora

IILM University, Greater Noida, U.P.

Roll No: 2410031349

Host Organization: Thiranex — Skill Development & Future Tech

Candidate & Program Overview

Student Name	Manthan Arora
University	IILM University, Greater Noida, U.P.
Degree Program	B.Tech — Computer Science & Engineering (IBM)
Roll Number	2410031349
Internship Domain	Data Science
Host Organization	Thiranex (Skill Development & Future Tech)
Duration	10 Aug 2026 – 09 Sept 2026
Certificate ID	THX-AUG1026-364



Internship Snapshot

4 Weeks

Intensive Duration

Virtual Mode

Remote Delivery

Verified

MSME-Registered Provider

Organization Profile: Thiranex

Thiranex is a skill-development and future-tech training organization focused on building industry-relevant, hands-on expertise among students and early-career professionals across emerging technology domains.



MSME Registered

Operates as a recognized entity under India's Ministry of Micro, Small & Medium Enterprises.



Mentor-Led Training

Internship programs guided by industry practitioners under the leadership of Founder & CEO Hariharan M.



Future-Tech Focus

Core training tracks span Data Science, AI, and emerging technology skill development.



Verified Certification

Each completion is issued a verifiable certificate with a unique QR-coded credential ID.

Problem Statement & Industry Context



The Industry Gap

Academic coursework introduces statistics and machine-learning theory, but most students graduate with limited hands-on exposure to real, messy datasets and end-to-end analytical workflows used in industry.

A model or insight has little business value until it is built on clean data, validated rigorously, and communicated through clear visual storytelling to decision-makers.



The Internship's Scope

- ◆ Sourcing, cleaning, and preparing real-world datasets
- ◆ Exploratory Data Analysis (EDA) to surface patterns & anomalies
- ◆ Building and evaluating machine-learning models
- ◆ Communicating findings through dashboards & visual reports

Core Internship Objectives



Data Wrangling

Collect, clean, and structure raw data using Python (Pandas, NumPy) for downstream analysis.



Exploratory Analysis

Apply statistical techniques and visualization to uncover trends, correlations, and outliers.



Machine Learning

Build and evaluate regression, classification, and clustering models with Scikit-learn.



Insight Communication

Translate analytical findings into dashboards and reports for non-technical stakeholders.

Tools & Technologies Used



Programming & Libraries

Python, Pandas, NumPy, Jupyter Notebook



Visualization

Matplotlib, Seaborn, Power BI



Machine Learning

Scikit-learn — Regression, Classification, Clustering



Data Handling

SQL, Excel, CSV/API data sources



Statistics

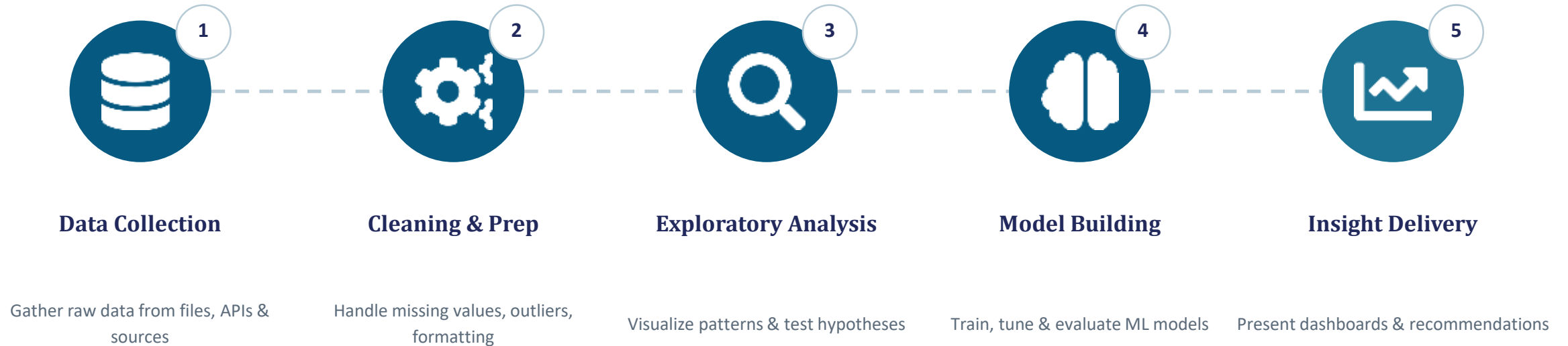
Descriptive & inferential statistics, hypothesis testing



Workflow

Data cleaning pipelines, feature engineering

Data Science Workflow Followed



This mirrors the standard industry data-science lifecycle — from raw data to a decision-ready insight — applied throughout the internship on practice datasets and assignments.

Internship Timeline (4 Weeks)

Week 1



Foundations & Data Handling

Python refresher, Pandas/NumPy, data collection, and cleaning fundamentals.

Week 2



Exploratory Data Analysis

Statistical summaries, correlation analysis, and visualization with Matplotlib/Seaborn.

Week 3



Machine Learning Modeling

Regression, classification & clustering with Scikit-learn; model evaluation metrics.

Week 4



Capstone & Reporting

End-to-end mini-project, dashboarding, and final presentation of insights.

Key Skills Demonstrated



Data Cleaning & Preprocessing

Handled missing values, duplicates, and inconsistent formats across raw datasets to prepare analysis-ready data.



Exploratory Data Analysis

Used descriptive statistics and visual analysis to identify trends, correlations, and outliers in datasets.



Predictive Modeling

Built and compared regression and classification models, evaluating performance using standard metrics.



Data Visualization & Reporting

Designed clear charts and summary dashboards to communicate findings to a non-technical audience.

Certification & Official Verification

Internship Title	Data Science Internship
Issuing Organization	Thiranex — Skill Development & Future Tech
Certificate ID	THX-AUG1026-364
Duration	10 Aug 2026 – 09 Sept 2026
Authorized Signatory	Hariharan M, Founder & CEO
Registration Status	MSME Registered



Verified Credential

This certificate carries a scannable QR verification code confirming authenticity against Thiranex's official records.

Internship Completion Certificate



Thank You

Manthan Arora

B.Tech CSE (IBM), IILM University, Greater Noida, U.P.

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